

GESTURE-CONTROLLED WHEELCHAIR

Using ATmega32

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Wheelchairs are useful for people for whom walking is difficult or impossible due to some illness, injury or disability. There are different types of wheelchairs. Manual wheelchairs are pushed using their handles. Motorised wheelchairs, are driven by joystick.

Voice-controlled wheelchairs are the latest development. These can be driven just by giving voice commands. A more advanced and intelligent version of the wheelchair is controlled directly through human mind, such as the one used by renowned scientist Stephen Hawking.

In case a person is unable to move the wheelchair even with joystick or voice command, an alternative is gesture-controlled wheelchair. The wheelchair moves as per user's finger gestures. The user has to simply bend his fingers to move the wheelchair.

Here we present the project to control a wheelchair through finger gestures. It uses flex sensors to record finger gestures. The gesture signals are fed to ATmega32 microcontroller, which processes the information to control four DC motors in order to move the wheelchair in any desired direction.

Flex sensors are widely used to convert finger gestures into equivalent electrical signals. These are actually variable resistors, whose resistance changes when these are bent in either direction. This change in resistance is converted into an equivalent voltage, which can be used for further processing. The author's prototype including the sensor arrangement is shown in Fig. 1. Flex sensor used by the author is shown in Fig. 2.

System block diagram

Block diagram for gesture control of a wheelchair is shown in Fig. 3. Major building blocks of the system are flex sensors, microcontroller ATmega32,

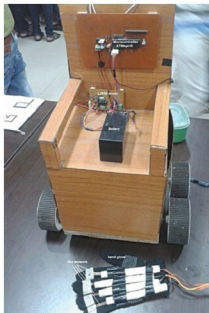


Fig. 1: Author's prototype including sensor arrangement attached to a hand glove



Fig. 2: Flex sensor

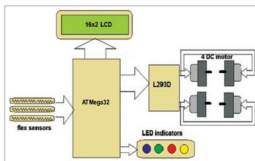


Fig. 3: Block diagram for gesture control of a wheelchair

motor driver L293D, DC motors, LCD display and LED indicators.

Flex sensor. It converts finger ges-

tures into analogue voltages, which are fed to the microcontroller.

ATmega32 microcontroller. It performs the following tasks:

1. Takes flex sensor input from the internal analogue-to-digital controller (ADC) and checks whether it exceeds certain threshold limit when the sensor is bent
2. Rotates right and left motors in forward or reverse direction to move the wheelchair in the desired direction
3. Displays wheelchair movement on the LCD and also displays the direction of the wheelchair through LED indicators

LCD. It displays messages about the wheelchair's movement direction as well as other messages

L293D driver. It provides sufficient voltage and current for DC motors to drive the wheelchair

DC motors. The four DC motors move the wheelchair in all the four directions: forward, reverse, left and right. Two motors (front and rear) on one side, say left side, are connected in parallel. Similarly, the other two motors on the right side of the wheelchair are connected in parallel. So there are four motors for four wheels, with one motor for each wheel.

LED indicators. These indicate the wheelchair's direction of movement (forward, reverse, left or right).

Circuit and working

Circuit diagram of the gesture-controlled wheelchair using ATmega32 is shown in Fig. 4. As mentioned earlier, it is built around four flex sensors (FS1 through FS4), an ATmega32 microcontroller (IC1), a 16x2 LCD (LCD1), an L293D motor driver (IC2), four 12V DC motors (M1 through M4) and a few other components.

Flex sensors (FS1 through FS4) are